Omkar Patil

Education

Indian Institute of Technology Bombay 2020–2024 B.Tech Mechanical Engineering (Minor in Electrical Engineering) CGPA: 8.2

- Class X (Maharashtra Board): 94.6Class XII (Maharashtra Board): 90.15
- JEE Advanced: 98.85 percentile (AIR 2879); JEE Mains: 99.44 percentile (AIR 6305)
- National Merit-cum-Means Scholarship (2016); State Merit Scholarships in multiple school years
- Relevant Coursework: Manufacturing Processes, Statistical ML and Data Mining, Microprocessors and Automatic Control

Work Experience

Bajaj Auto Ltd. Aug 2024–Present *Assistant Manager, RD Testing*

- Led execution of 160–180 durability tests annually while managing a 7-member team.
- Designed and deployed IoT-enabled monitoring and auto-stop system across 6+ rigs, saving ∼3000 hours annually and improving lab throughput by 5–6
- Implemented abnormal stoppage alerts to reduce silent failures and rework.
- Automated reporting workflows, saving 20 man-hours across 30+ reports.
- Collaborated with design and CAE teams to diagnose and resolve 8+ field failures.
- Standardized pre-test preparation and introduced ownership-based execution model.
- Led safety improvement initiative in impact testing setups; received Company-level Safety Kaizen Award.

Relevant Projects Industrial Exposure

Safety and Productivity Improvement, Tata Steel

- Identified automation interventions to reduce man-machine interface risks.
- Projected 20

Autonomous Underwater Systems (IITB + LT Defense)

- Designed subsea camera hull for 50m underwater operation with manufacturability considerations.
- Performed CFD drag analysis and optimized hull components using ANSYS.
- Led fabrication and mentored 5 members selected from 150+ applicants.

Inventory Management System

- Reduced inventory cost by 13.5
- Applied K-means clustering to streamline review cycles.

Cooling Channel Optimization in Injection Molding

- Applied Genetic Algorithm (GA) to optimize cooling-channel geometry for improved thermal performance.
- Applied Particle Swarm Optimization (PSO) to reduce injection molding cycle time through improved channel design.

Academic Responsibilities

Teaching Assistant, IIT Bombay

- Supported 40+ students in Engineering Graphics and lab activities.

Technical Mentor, Institute Technical Summer Project

- Guided a team of 4 students in a technical project involving mechanical design and prototyping.

Skills

Programming: Python, C/C++ **Systems/Tools:** ESP32, Flask, Streamlit, ANSYS, SolidWorks

Extracurricular Interests

- Completed year-long National Cadet Corps (NCC) program at IIT Bombay.
- Completed snow trek to Kedarkantha summit (3810m) under winter conditions.
- Undertook 83m bungee jump and 16 km river rafting in Rishikesh.
- Interests: Chess, Sketching, Gardening.